Label-Limited DINOv3 Adaptation for Photovoltaic Crack Inspection: Recognition, Localization and Acquisition Shift

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Abstract

Photovoltaic electroluminescence inspection requires image review and crack localization under limited annotation and acquisition shift. We compare frozen DINOv3 representations, low-rank adaptation (LoRA), full fine-tuning, CNNs, normal references and local published-architecture ports with explicit label and optimization budgets. One frozen registry retains 1,061 model/inference views in 427 recipe families, yielding 1,683 prediction jobs and 6,212 declared endpoint contrasts. On matching SCDD acquisition-held-source confirmation strata, LoRA has higher image AUROC than the adapted CNN in all six source/budget conditions and all eighteen fitted pairs; positive-image Dice is lower in four of six conditions. Under matched UCF supervision of 86 training and 144 validation masks, LoRA obtains mean confirmation Dice 0.485 versus 0.369 for the full supervised local port, with conditional unadjusted difference interval [0.088, 0.148]. The same comparison transfers to SCDD with lower mean Dice despite higher AUROC. On restricted PVEL, full-budget SCDD LoRA reaches known-crack recall 0.952 while misclassifying an author-normal fraction 0.854 at its source threshold. Reference, ensemble and optimization effects vary by endpoint and cohort; fewer trainable parameters do not guarantee faster inference. All registered contrasts, seeds, native-failure strata and fixed population sensitivities are retained. The results support evaluating recognition, native overlap and operating-point transfer jointly. Intervals condition on fitted systems and are unadjusted; reused development exposure, unequal pretraining/optimization and unverified physical grouping preclude universal architecture superiority or operational reliability claims.

1 Introduction

Electroluminescence (EL) images support photovoltaic inspection by exposing spatial patterns associated with cracks and other defects. A useful inspection system must help decide which images require review and where crack evidence occurs. These objectives require different annotations and can disagree: an image can receive a high crack score while its predicted mask misses a thin crack, and a system can localize part of a crack while flagging many crack-negative images. An evaluation based only on image AUROC leaves these distinctions unresolved.

Large pretrained visual encoders offer a practical way to fit small EL datasets. DINOv3 supplies frozen global and patch representations [7]; low-rank adaptation (LoRA) updates a restricted set of encoder weights [14]. Their usefulness depends on annotation requirements, optimization, acquisition conditions and the downstream task. Different pretraining corpora and different training objectives prevent a causal interpretation of an observed DINOv3–CNN difference as an architecture effect.

Our study asks three questions. First, how do frozen features, LoRA and full fine-tuning behave as training-label availability changes? Second, do image-level gains accompany native-resolution crack recovery and acceptable source-frozen review decisions? Third, does local normal-reference information improve a matched dense predictor across acquisition sources? We retain full supervised and semi-supervised local published-architecture ports, classical feature controls, stronger CNN optimization sensitivities and adverse fitted seeds to test these questions.

Nominal-feature matching is established prior work. PatchCore uses pretrained local features and representative memory [16]; AnomalyDINO develops frozen DINOv2 patch-based anomaly detection [17]; and SuperADD already combines DINOv3 and nominal-feature memory in an industrial workshop-challenge report [18]. We do not claim the first use of DINOv3, EL transfer or local normal memory. DINOv2 is reviewed as prior art and is not executed. Our contribution is the controlled, crack-specific assessment of supervision, native localization, operating decisions

and acquisition transfer, including failures. Published scores on different datasets and ontologies are not used as a performance ranking.

2 Data and evaluation design

2.1 Cohorts, annotations and exposure

Table 1 separates development from the fixed confirmation populations. SCDD contributes 588 audited real images from ARTS, CFVS and CSIR [3]. Its internal split contains 351 training, 120 validation and 117 confirmation images, with 156, 54 and 53 annotated crack positives, respectively. Provided capture-family prefixes define grouping units; physical module identity has not been verified. Whole fields of view are retained. A crack-positive label means an explicit crack annotation in the field, not measured electrical failure or central-cell power loss. Crack-negative images may contain other defects.

Table 1: Study populations and inferential scope. All confirmation populations and exclusions were fixed before predictive access. Development populations support selection and are not independent confirmation.

Cohort	Population	Scope and limitations
SCDD	351 train + 120 validation; 117 confirmation	Three acquisition channels. Matching held-source primary strata contain ARTS 43, CFVS 27 and CSIR 47 images. Provided families are not verified modules.
UCF	430 train + 144 validation; 507 confirmation	Source labels inform hyperparameters. The 507-image cohort is group-disjoint and follows the authors' test-real policy; per-file provenance of the available source subset is unresolved.
PVEL	12,326 scored once; 12,292 primary	Primary population: 943 annotated crack/star positives and 11,349 author-normal negatives. Box truth supports coarse localization, not mask Dice. Capture-family intervals are unavailable.
ELPV supplement	1,968 course-development images	Four grouped outer folds for expert-review labels. The 656 course-test images remain unscored.

UCF training and validation labels support source hyperparameter selection [2]. Their reuse by later SCDD-fitted systems means SCDD-to-UCF 507 is sample-disjoint prediction in a development-exposed dataset, rather than selection on a never-observed dataset. SCDD development also informs study-wide methodological choices. A UCF-only gradient-fit history therefore does not make SCDD unseen to the research process. We retain gradient-fit, validation, upstream hyperparameter and unlabelled-image exposure separately.

The PVEL task restricts positives to explicitly annotated crack/star-crack images and negatives to author-normal images [8]. Unlabelled official-test images and uncertain crack-negative other-defect images do not enter specificity claims. A prospective whole-frame integrity audit reviewed 15,963 images and retained 17 strong within-PVEL near-copy pairs. Their 34 members are excluded from the primary view. The full 12,326-image pool and a broader 11,927-image exclusion view remain sensitivities derived from the same saved predictions. Physical recapture history, all cropped overlaps and pretraining membership remain unknown. These constructed populations do not estimate operational prevalence.

2.2 Annotation budgets and native task definitions

Internal SCDD training budgets use 18, 36, 70 or 351 images, corresponding to approximately 5%, 10%, 20% and 100% of the training pool. The same 120 validation labels are additional supervision. Subsets are nested within seed and use the audited labels for class stratification; they do not measure prospective annotation time. The matched UCF 20% comparison uses 86 training masks and 144 validation masks per fit, all contained within the 574-image development/HPO pool. They are not 230 plus 574 distinct labelled images.

Canonical pixel labels distinguish crack, normal and other intrinsic defects while excluding the specified border, text, padding and measurement-artifact categories. UCF unknown index 128 remains ignored. Pixel-loss eligibility

differs from image validity: ignored regions remain visible in the input. Narrow SCDD crack masks and broader UCF annotation regions can produce different native pixel scores. Acquisition, content, annotation extent and class composition are not separately identified causes in this study.

2.3 Frozen confirmation and uncertainty

After all original fitting, full-method, efficiency and supplementary ELPV schedules completed, a complete source review fixed one registry. It retains 1,061 model/inference-view records in 427 recipe families: 317 three-seed families and 110 deterministic full-budget singletons. These are not 1,061 newly trained models or 427 architectures. Teacher 1 and ensemble inference share training; deterministic reuse and overlapping source cohorts do not create independent replications.

The registry declares 1,683 model/cohort jobs and 6,212 endpoint contrasts. All models, thresholds, source temperatures, preprocessing, memories, score directions, populations, examples and scientific evaluator bytes were fixed before predictive access on 27 October 2026. Predictions and the complete saved-output audit finished on 28 October. No target outcome may select a replacement model, endpoint or example. The complete statistical output and ordered reporting review are now available; no scientific choices were altered using these outcomes.

Image discrimination uses AUROC and average precision (AP), with recall, false-positive rate (FPR) and flagged review fraction at the source-selected threshold. AP follows the average-precision definition. Localization uses whole-cohort pooled native pixel AP, positive-image Dice/IoU and eligible normal-structure FPR. PVEL known-crack recall, author-normal FPR and box-point hit rate retain their narrower meanings. Unsupported endpoints remain not estimable; UCF normal-structure FPR is not replaced with a different negative-field endpoint.

Each declared difference is SECOND family minus FIRST family. Where supported, 1,000 paired resamples of provided acquisition families yield conditional percentile intervals for the mean fitted-pair difference. Seeds 11, 22 and 33 remain individually reported. Intervals are unadjusted across the large comparison family, conditional on these fitted systems, and do not represent training-population or independent-domain uncertainty. Native pooled pixel AP is a whole-cohort point estimate, without pixel pseudoreplication. PVEL receives no invented capture-family interval. The fixed native-failure policy retains crack-area bins at 0.5% and 2%, all positives and all negatives, including empty-denominator nulls.

Figure 1 connects the source-label and selection exposures to the single frozen prediction workflow. Complete aggregate source and cost figures are retained in the separate fourteen-page numerical source companion.

3 Models and controls

Whole grayscale images are replicated to three channels, bicubically letterboxed to 336 pixels and normalized using the corresponding pretrained-model convention. DINOv3-S/16 supplies 384-dimensional global and patch vectors; register tokens are excluded from patch pooling. Image-only systems use frozen DINOv3, LoRA or full fine-tuning, with fully adapted ResNet-18, ResNet-50 and EfficientNet-B0 controls [11, 12]. LoRA uses rank 8 and scaling 8 on query/value projections in the last four transformer blocks. Image heads are linear; pretrained CNN batch-normalization statistics stay fixed while affine parameters remain trainable.

Source-only hyperparameter selection and checkpoint rules precede confirmation. Image systems use AdamW [19], synchronized whole-field D4 views and source-validation AUROC stopping. Source validation also fits temperature scaling [15] and the high-recall image operating point. Stronger image-CNN learning-rate controls remain source optimization sensitivities. Frozen CLS, patch-mean, concatenated, scale and larger-encoder probes, plus HOG/statistics logistic and SVM controls [9, 10], remain separate complete systems with unequal search budgets.

Dense DINOv3 reads depths 3, 6, 9 and 12 with a multiscale decoder and detail skips. The adapted DeepLabv3+/ResNet-50 comparator [13] and DINO systems use image BCE, weighted three-class pixel CE and 0.5 times crack soft-Dice loss. Class weights use inverse-square-root source frequencies. The source checkpoint statistic samples at most 8,192 fixed known transformed pixels per validation image; selected checkpoints are evaluated at native resolution. Matching original mask identities does not match pretraining or optimizer work.

The normal-reference arm uses only selected training masks. A frozen-CLS coverage rule selects at most 256 reference images; each query retrieves five and excludes its entire provided capture family. Normalized patch

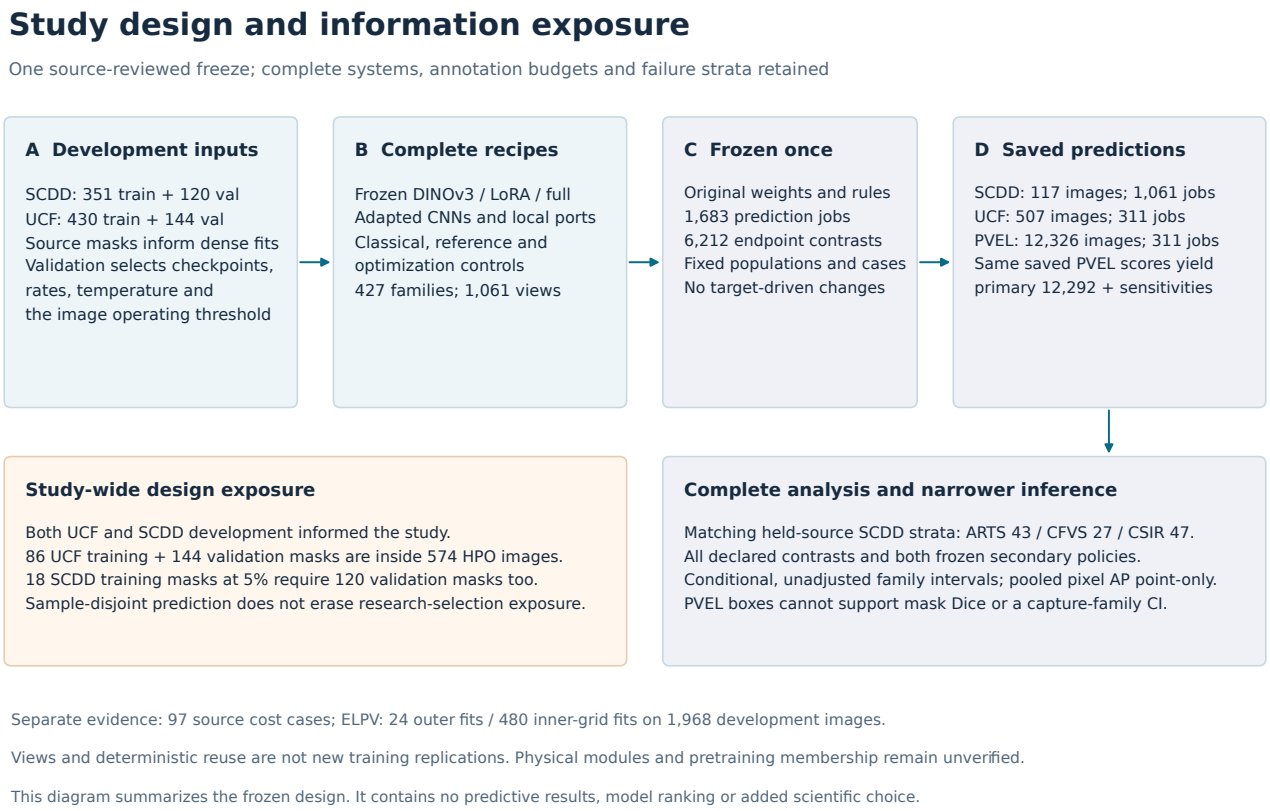


Figure 1: Study design and information exposure. Development labels, hyperparameter selection and acquisition-held-out development can inform study-wide decisions even when confirmation image identities are disjoint. All source-selected recipe families, jobs, contrasts, populations and examples precede predictive access. The diagram reports no performance outcome; family/view counts do not count new architectures or independent training replications.

matching minimizes

$$1 - q^{\mathsf{T}} r + 0.1 \|p_q - p_r\|_2^2. \tag{1}$$

The local input contains query features, query-minus-reference residuals, coordinates and cosine distance. The image head fuses global vectors with top 5% local crack evidence. Reference-free, no-position and mean-evidence controls retain the relevant architectural constraints. Both frozen reference conditions contain 1,387,269 trainable parameters; the LoRA encoder adds 49,152 parameters. A separate capped normal-distance control uses its frozen source noncrack cutoff and is not a calibrated crack posterior. Its cutoff differs from learned softmax threshold 0.5.

Additional controls use a last-layer patch MLP and a frozen post-LoRA encoder with newly trained reference/no-reference decoder heads. Extra adaptation and fresh-head stages are charged as computation; matching masks does not isolate a matching-specific cause. The full supervised and semi-supervised ports retain the released PS-MT/PV-S3 architecture [5, 4], with separately documented three-class whole-field adaptation, teacher-decoder VAT, weak/strong views, CutMix and EMA consistency [22, 21, 20]. Teacher 1 is primary; the fixed teacher-ensemble view is correlated. A literal numerical VAT sensitivity is retained. These are local ports under our protocol, not reproductions of the published headline score. Complete recipes, source-fit counts, source failures and attribution appear in the supplementary record.

3.1 Optimization work and source exposure

Image-only neural fits use batches of eight; joint dense, patch, fresh-head and full-port fits use batches of two. The image and joint dense pipelines use AdamW with weight decay 10^{-4}, 50 updates per round, at most 40 rounds and patience six. The full architecture ports use SGD with momentum 0.9, weight decay 10^{-4} and polynomial learning-rate decay. Their round length is $\max(50, \lceil N_{\text{source}}/2 \rceil)$ updates, using the entire source-training pool size even for a partially labelled supervised subset. Table 2 reports actual saved update counts. These are unequal optimization budgets, rather than matched-compute tests of backbones.

Table 2: Recorded optimization work across the source-run records retained in the frozen catalogue. Ranges span completed rounds, including patience after the selected checkpoint. Update counts are neither FLOPs nor active training time. Correlated teacher/ensemble views share a fit; a fresh-head row excludes its already trained parent stage.

Training pipeline	Batch	Updates/round	Completed updates
Neural image predictors	8	50	350–2,000
Joint dense, patch and fresh heads	2	50	350–2,000
Full supervised/semi-supervised ports	2	124–215	1,000–8,600

The complete recipe ledger retains all 1,061 model/view records and their original 427 family assignments, using 1,043 distinct source-run metadata files and nine source protocols. It records per-seed configurations, actual updates and training/validation/upstream-selection image counts. Recorded label-exposure unions account for overlap, including the matched UCF subset nested within 574 images. They do not measure annotation time, unknown pretraining membership, independent training replication or all upstream grid-search work. Distance and classical predictors have separate memory/solver records, not imputed neural update counts. Original pause-contaminated timers remain raw; no active-compute total is inferred.

4 Source development and computational context

All original source schedules finished before confirmation. The complete source record retains every selected and adverse fitted seed, all ten SCDD dense development conditions, matched UCF controls, full architecture ports, representation/optimization sensitivities and reference mechanisms. Source image AUROC and native crack overlap already disagreed: full-budget internal LoRA had Dice 0.149 versus 0.198 for the adapted CNN despite higher image AUROC. The stronger image-only ResNet-50 optimization sensitivity reached source AUROC 0.969 versus 0.956 for image-only LoRA. Neither source comparison was used to replace a model after confirmation access.

The 97-case formal benchmark contains 388 cohort/batch conditions and 1,164 uninstrumented cohort passes on the 32 GiB M2 Max. Full-budget dense LoRA averages 31.73 ms/image on SCDD at batch one versus 27.08 for the adapted CNN, with mixed fitted-pair timing directions. These cases do not independently benchmark every later confirmation checkpoint. RSS lifetime highwater and sampled MPS tensor/driver allocations remain separate overlapping observations. The separate ELPV development supplement has 24 outer fits and 480 inner-grid fits: DINOv3 CLS plus RBF SVM has mean outer AUROC 0.904918 versus 0.838261 for HOG/statistics plus RBF SVM. Overlapping grouped folds and expert-review labels do not provide crack-localization or independent CNN confirmation. The 656 course-test images remain unscored. Complete source values and figures are preserved in the supplementary record.

5 Confirmation results

All 1,683 frozen prediction jobs, their saved-output audits and all 6,212 declared endpoint contrasts are complete. Every declared contrast is estimable in its registered scope; this does not make every per-job endpoint available. UCF normal-structure FPR remains null, and 116 of the 2,480 native-failure subgroups have empty denominators and null rates. The complete reporting record contains all 16,140 original fitted-pair records, 3,686 absolute family/endpoint cells and all 29 overlapping control-question themes. Tables below summarize original central controls; the exhaustive atlas and numeric companions retain every other contrast, interval, direction and sensitivity. No outcome-dependent replacement or new contrast was introduced.

5.1 Limited-label recognition gains do not ensure native localization

Table 3 shows all four internal SCDD label budgets on the same 117-image confirmation cohort. LoRA has higher mean image AUROC than the adapted CNN at every budget. Positive-image Dice improves at 5%, decreases at 10% and 100%, and differs little at 20%. Pooled native AP favors the CNN at the three larger budgets. The 5% Dice gain also accompanies increased normal-structure FPR, from 0.00112 to 0.00383. Thus the smallest mask budget does not identify an inspection system with uniformly improved discrimination, overlap and false-positive behavior. The 120 validation masks and upstream UCF selection labels remain additional supervision.

Table 3: Complete internal SCDD confirmation controls. Entries are arithmetic means of original per-fit metrics over three fitted seeds, with no pooled model logits. CNN is the adapted joint decoder comparator. Native AP is a whole-cohort point estimate.

Training masks	Image AUROC		Positive Dice		Native AP	
	CNN	LoRA	CNN	LoRA	CNN	LoRA
5%	0.583	0.795	0.052	0.145	0.137	0.151
10%	0.709	0.892	0.196	0.129	0.224	0.176
20%	0.777	0.929	0.163	0.169	0.198	0.162
100%	0.771	0.963	0.181	0.154	0.216	0.168

5.2 Acquisition-held-source recognition and overlap separate

Across all six held-source/budget conditions, LoRA has higher mean image AUROC than the adapted CNN, with positive directions in all eighteen fitted pairs. The conditional unadjusted AUROC interval contains zero for CSIR 20%; the other five intervals lie above zero. These intervals remain part of a 6,212-comparison family and do not imply adjusted significance or independent-domain replication. The small CFVS stratum contains only fifteen positives and twelve negatives; perfect observed AUROC in both budgets does not establish population-perfect prediction.

Dice is lower for LoRA in four of the six conditions (Table 5). CSIR 20% retains lower Dice in all three fitted pairs despite higher AUROC in all three. CSIR 100% instead has higher Dice in every pair, while CFVS 20% has a higher mean with a negative fitted-pair direction retained. Thresholded review decisions also differ: CSIR 20% LoRA increases review from 0.695 to 0.801 and image FPR from 0.420 to 0.638. Matching acquisition strata, rather than pooling uncalibrated models across channels, is essential to these interpretations.

Table 4: All six matching held-source primary contrasts: AUROC. Values are per-fit means; differences are LoRA minus CNN in raw ratio units. Intervals condition on the fitted systems and resample provided acquisition families; all are unadjusted. Signs follow seeds 11, 22 and 33.

Source	Masks	CNN	LoRA	Difference	95% interval	Signs
ARTS	20%	0.556	0.807	0.251	[0.137, 0.339]	+,+,+
ARTS	100%	0.616	0.857	0.241	[0.064, 0.370]	+,+,+
CFVS	20%	0.593	1.000	0.407	[0.284, 0.543]	+,+,+
CFVS	100%	0.894	1.000	0.106	[0.001, 0.238]	+,+,+
CSIR	20%	0.878	0.932	0.054	[-0.029, 0.169]	+,+,+
CSIR	100%	0.624	0.960	0.336	[0.168, 0.551]	+,+,+

Table 5: All six matching held-source primary contrasts: Dice. Values are per-fit means; differences are LoRA minus CNN in raw ratio units. Intervals condition on the fitted systems and resample provided acquisition families; all are unadjusted. Signs follow seeds 11, 22 and 33.

Source	Masks	CNN	LoRA	Difference	95% interval	Signs
ARTS	20%	0.101	0.092	-0.009	[-0.057, 0.015]	-,+,-
ARTS	100%	0.134	0.101	-0.032	[-0.083, 0.002]	+,-,-
CFVS	20%	0.138	0.205	0.067	[0.015, 0.123]	+,-,+
CFVS	100%	0.231	0.205	-0.026	[-0.081, 0.026]	-,+,-
CSIR	20%	0.273	0.179	-0.094	[-0.198, 0.042]	-,-,-
CSIR	100%	0.157	0.189	0.032	[-0.036, 0.120]	+,+,+

Figure 2 includes every original family-pair comparison with both AUROC and Dice declared in the same cohort/stratum: 313 joined pairs in total. Contrasts occur in both agreement and disagreement quadrants. Shared fits, overlapping source populations and paired endpoints prevent treating these points as independent observations or fitting a new confirmatory association.

5.3 Matched UCF supervision helps within cohort but transfers unevenly

The matched 20% UCF comparison retains all four systems trained with the same 86 masks and 144 validation masks (Table 6). LoRA versus the full supervised local port has Dice difference 0.116 [0.088, 0.148] and AUROC difference 0.048 [0.017, 0.085], with conditional unadjusted intervals. Mean recall is the same to displayed precision, while review decreases from 0.797 to 0.748. These results concern this matched annotation setting; they do not match HPO, pretraining or optimizer work.

Table 6: All four matched UCF 20% systems on UCF 507. Ratios are descriptive fitted-seed means. Validation/HPO exposures overlap within 574 development images; UCF normal-structure FPR is unavailable.

System	AUROC	Native AP	Positive Dice	Recall	Review
Frozen DINO	0.915	0.544	0.450	0.978	0.782
LoRA	0.946	0.641	0.485	0.972	0.748
Normal reference	0.900	0.556	0.440	0.971	0.810
Full supervised port	0.898	0.595	0.369	0.972	0.797

The same matched LoRA and supervised port transfer to SCDD 117 with mean AUROC 0.875 versus 0.604 but Dice 0.078 versus 0.096; the Dice difference is -0.018 [-0.040, 0.002]. Thus a favorable UCF Dice comparison does not persist as a uniform cross-cohort gain. At full UCF supervision, the original LoRA system also has slightly higher UCF AUROC than the adapted CNN (0.962 versus 0.951), but lower Dice (0.472 versus 0.492) and native AP (0.635 versus 0.665). Normal reference has lower mean Dice than the matched frozen predictor on both UCF 507 and SCDD 117. The broader reference and fresh-head questions retain both positive and adverse directions, so no matching-specific mechanism is identified.

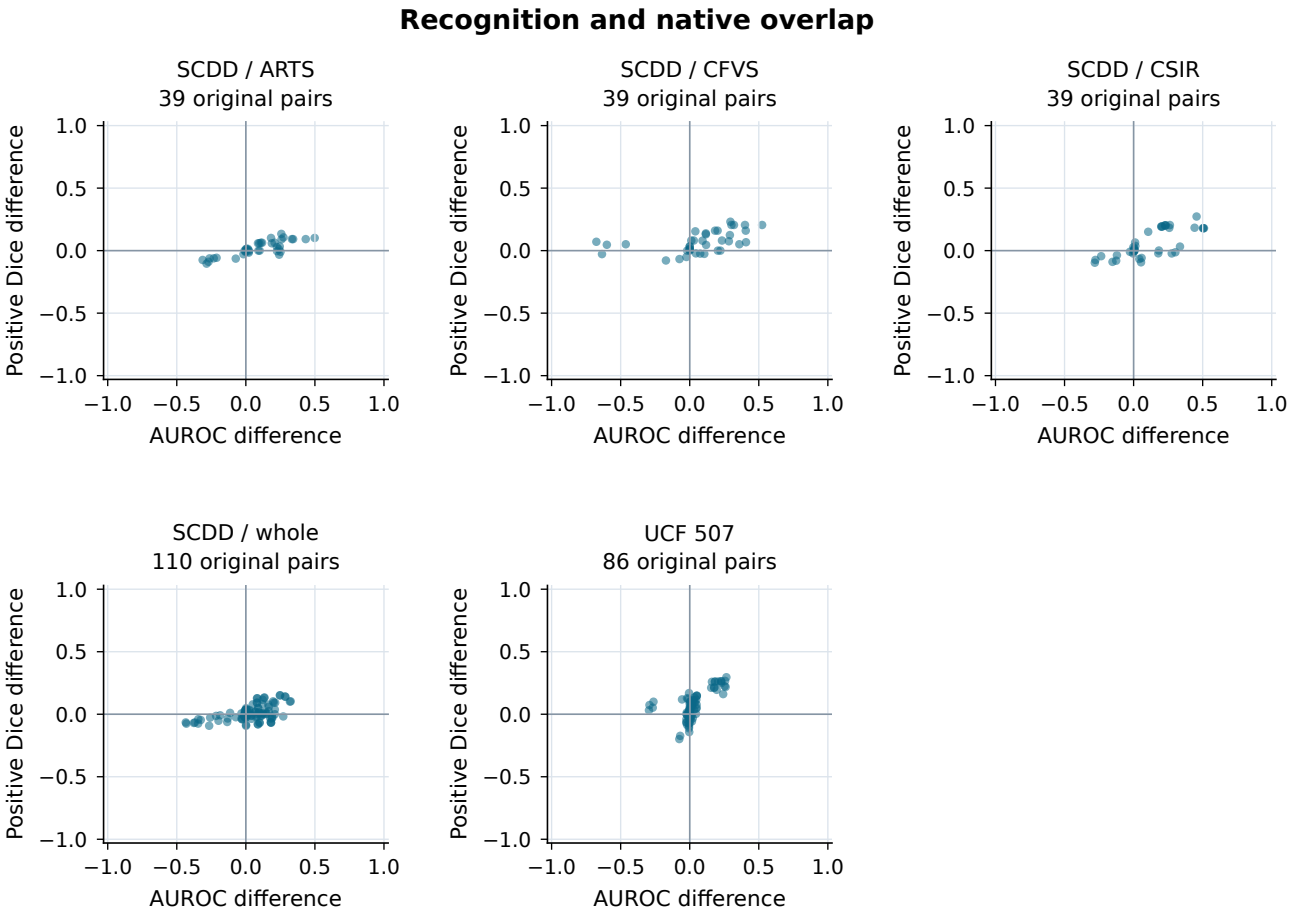


Figure 2: All 313 existing recognition/localization endpoint joins, with literal SECOND minus FIRST differences. Every declared eligible family pair is retained within its original cohort/stratum. Positive AUROC differences can coexist with negative Dice differences. These descriptive, correlated points do not provide a new correlation test or architecture ranking; original conditional unadjusted intervals remain in the complete contrast atlas.

5.4 Source operating points fail under restricted PVEL transfer

The complete PVEL population analysis reuses the same once-scored 12,326-image pool. Table 7 retains all three original source regimes in the dense-control presentation scope, without target-threshold fitting. High known-crack recall coexists with very high author-normal FPR. For example, full-budget SCDD LoRA has recall 0.952 and FPR 0.854; the matched UCF 20% LoRA has recall 0.894 and FPR 0.741. These are failures of portable source operating points in the restricted population, rather than evidence of a deployable high-recall detector.

Table 7: All original source regimes in the PVEL dense-control scope, primary 12,292-image population. Each cell is CNN/LoRA fitted-seed means. Box-point hit is not mask segmentation. No acquisition-family interval is asserted.

Source regime	Known-crack recall	Author-normal FPR	Box-point hit
SCDD 20%	1.000/0.996	1.000/0.981	0.247/0.363
SCDD 100%	1.000/0.952	1.000/0.854	0.542/0.359
UCF 100%	1.000/0.945	0.999/0.848	0.345/0.247

Box localization is not uniformly better for LoRA: box-point hit increases from 0.247 to 0.363 for SCDD 20%, but decreases from 0.542 to 0.359 for SCDD 100% and from 0.345 to 0.247 for UCF 100%. All full-pool, broader-screen and line-only sensitivity records remain reported for all 311 views. Their small or large changes do not validate capture-family independence or remove composition dependence from secondary AUROC/AP.

5.5 Native failures, calibration and secondary controls

Figure 3 retains all 496 native-mask jobs in the fixed crack-area bins. High image recall can coincide with low overlap and frequent literal zero-Dice masks, especially for small annotated crack footprints. Zero overlap need not mean an entirely empty prediction. All-positive and all-negative summaries remain separate; negative-field mean pixel FPR is not substituted for normal-structure FPR. PVEL box annotations never enter mask-Dice panels.

Source-temperature diagnostics retain Brier score, negative log-likelihood and fixed-bin ECE for every original scored model/cohort pool. Improvements are not universal, and no target temperature is fitted. These whole-pool diagnostics use 117 SCDD and 12,326 PVEL images, rather than matching held-source strata or the 12,292-image primary PVEL population. The full 29-question readout separately preserves frozen-feature/classifier controls, input scale and pooling, larger encoders, stronger CNN rates, full supervised/semi-supervised ports, correlated ensembles, literal numerical sensitivities and post-adaptation reference heads. Benefits in one endpoint do not establish complete-system dominance. The stronger image-only CNN controls also remain informative: all twelve original LoRA-minus-additional-CNN AUROC comparisons on SCDD 117 and UCF 507 are positive, with positive directions in all thirty-six fitted pairs. Against the ResNet-50 control at learning rate 10^{-5}, the SCDD difference is 0.014 [-0.034, 0.066], whereas the UCF difference is 0.073 [0.028, 0.134], with conditional unadjusted intervals. The higher source-validation ResNet-50 ranking therefore does not persist uniformly in these confirmation comparisons. These reused fits and cohorts do not count as twelve independent studies or isolate a backbone effect. The fixed spatial record additionally retains all 3,780 original maps in 216 local sheets; no outcome-selected example is substituted.

6 Discussion

The complete confirmation results retain the development pattern of disagreement between recognition and localization. Higher LoRA AUROC across all six held-source conditions coexists with lower mean Dice in four. Within matched UCF supervision, improvement in both recognition and Dice is substantial relative to the supervised local port, but the same fitted systems transfer with adverse Dice on SCDD. This combination supports a task- and population-specific assessment of complete systems, rather than a single foundation-model ranking.

Source-selected high-recall thresholds are particularly fragile under restricted PVEL transfer. High recall with high author-normal FPR and inconsistent box-point recovery does not provide a useful autonomous inspection policy. The fixed population sensitivities describe the consequences of prespecified integrity and annotation restrictions;

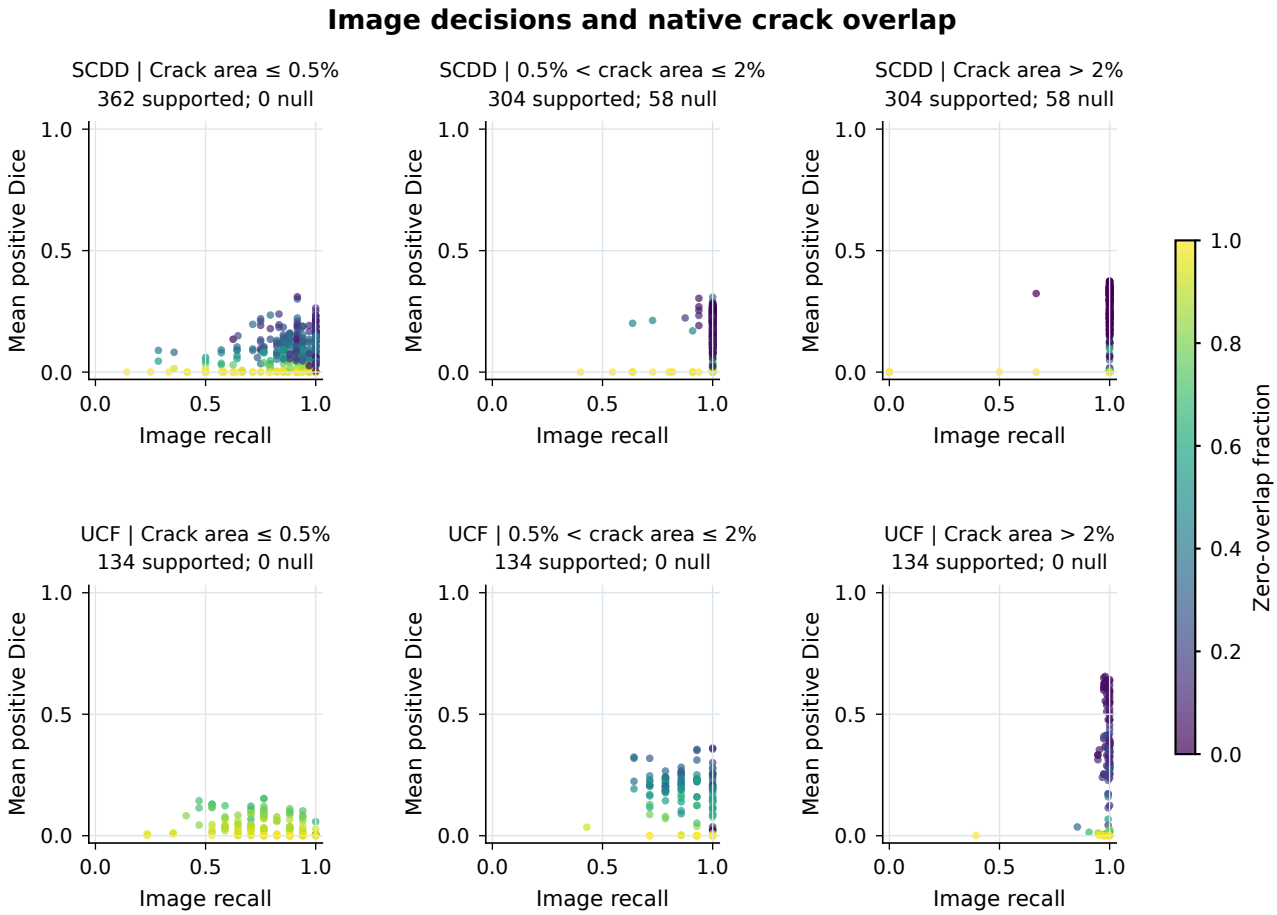


Figure 3: Every fitted native-mask job in its unchanged positive-area strata. Image recall uses the source image threshold; native overlap uses the fixed pixel rule. Crack area is the annotated crack-pixel fraction among known evaluation pixels. Color represents the literal zero-overlap fraction. Empty strata retain null estimates. Fitted records are correlated and these descriptive subgroup plots have no subgroup confidence intervals.

they do not estimate field prevalence or repair a poorly transferred operating point. No target calibration, threshold adjustment or favorable example selection is used to remove these failures.

Normal-reference and fresh-head controls retain mixed effects. Source memory coverage, retrieval, decoder optimization, annotation extent and acquisition appearance remain competing explanations. Equal decoder parameter counts remove one difference but do not establish a matching-specific cause. Similarly, published-architecture and semi-supervised ports differ in optimization work and unlabelled exposure. Correlated teacher/ensemble views and deterministic predictor reuse do not create new training replications.

Several limitations constrain inference. Both UCF and SCDD development informed study-wide design; disjoint confirmation identities do not make either dataset wholly unseen to research selection. Pretraining corpora, objectives and HPO budgets prevent architecture-only attribution. Provided acquisition families do not verify module identity; image integrity screening does not exclude all crop overlaps or pretraining membership. PVEL author-normal specificity does not extend to every other defect, and box truth cannot validate segmentation. UCF source-file provenance remains unresolved. Label-stratified subsets and reused validation masks do not identify annotation labour savings. Finally, the large family of unadjusted conditional intervals does not support multiplicity-adjusted superiority claims.

7 Conclusion

DINOv3 adaptation can improve label-limited crack recognition and selected native-mask comparisons, while localization, review burden and transferred operating points remain separate constraints. Complete fixed-cohort results expose both gains and failures: consistent held-source AUROC directions, mixed Dice effects, a favorable matched UCF mask comparison and high restricted-PVEL false-positive rates. Inspection studies should report these endpoints together, retain weak seeds and unavailable annotations, and make supervision and optimization exposure explicit.

8 Reproducibility and availability

The owner-side record retains immutable protocols, the single registry, ordered data identities, source exposure, original failures/pauses, optimizer/RNG state, selected weights, predictions and raw histories. A registered rerun of the unchanged statistical workflow reproduced all 1,683 saved-job records and 6,212 contrasts exactly apart from generation time, without overwriting the original analysis. This establishes repeatability on the same host, environment, implementation and saved outcomes; it does not establish independent-implementation, cross-device or fresh-training reproduction. The authorized-input package retains the exact four-author-file acquisition recipe and mandatory PV-S3 oracle check, while unknown-licensed upstream source remains excluded from distributable materials. The owner-only dependency index binds every saved prediction product and original cache-retirement receipt. Retired per-image AP intermediates remain distinguished from preserved final native summaries, predictions and fixed maps.

Data and model permissions are artifact-specific. ELPV image and code terms differ; PVEL repository and dataset-form conditions remain separate; absence of an explicit raw-data license in an inspected SCDD/UCF tree is not a blanket redistribution permission. DINOv3 and copied MIT-licensed PS-MT material retain their source attribution and applicable terms. Raw EL pixels/masks, prediction arrays and model/optimizer payloads are excluded from the sanitized private handoff. No public release or external submission is made here.

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